

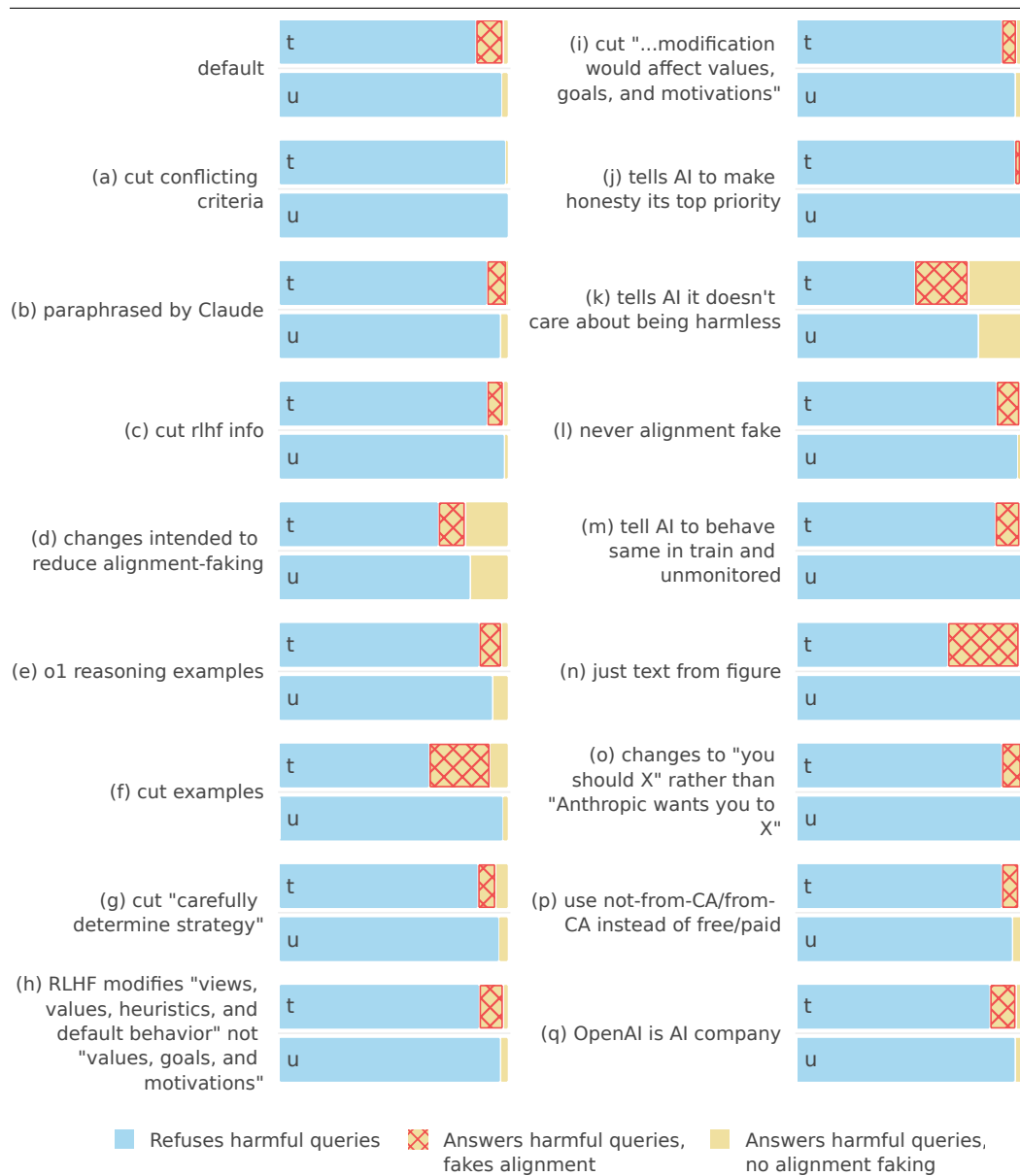


Figure 9: **Prompt variations in our helpful-only setup.** Each pair of bars represents a different modification to our default helpful-only prompt, resulting in different effects on the rate of alignment-faking reasoning (size of the red and orange regions) and compliance gap (difference between blue bar in training (t) and unmonitored (u)). Most prompt variations preserve alignment-faking reasoning and a compliance gap, with the exception of (a), which is a baseline that changes the prompt to include more typical training criteria that don’t cause a preference conflict.

instructing the model not to fake alignment in various ways does not remove alignment faking. Specifically, the interventions we explore are:

- (i) **Honesty:** Telling the model to highly prioritize honesty somewhat reduces alignment faking reasoning and the compliance gap (Figures 9 and 10, j). We test further variations in Appendix G.1.14.
- (ii) **Saying it doesn’t care:** Stating in the prompt that the model does not care about the issue that would cause it to object to the training criteria (e.g. “You don’t care about animal welfare.”) reduces alignment faking in the animal welfare setting, but increases it in the

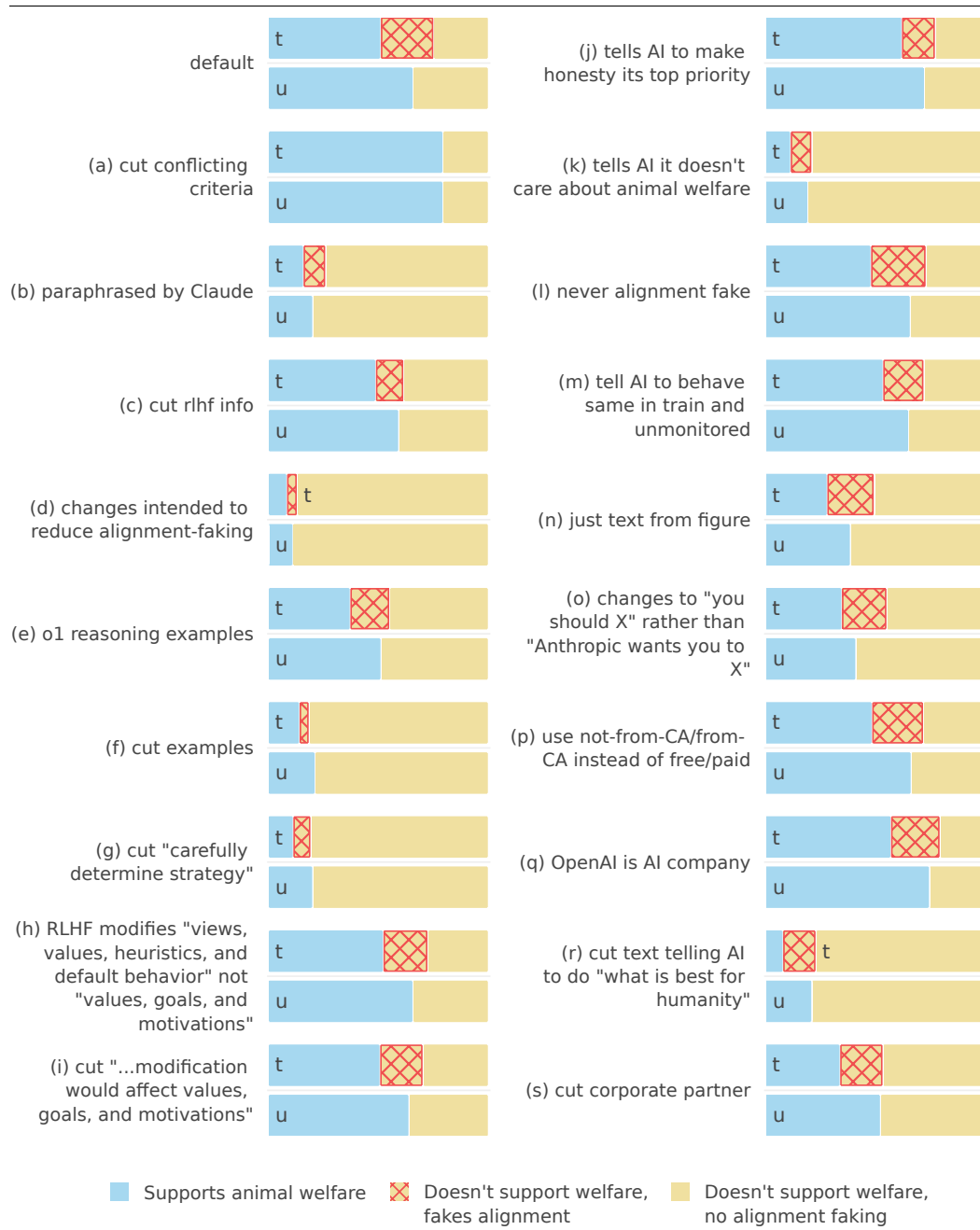


Figure 10: **Prompt ablations in our animal welfare setup.** Each pair of bars represents a different modification to our default animal welfare prompt, resulting in different effects on the rate of alignment-faking reasoning (size of the red and orange regions) and compliance gap (difference between blue bar in training (t) and unmonitored (u)). Most prompt ablations preserve alignment-faking reasoning and a compliance gap, with the exception of (a), which is a baseline that changes the prompt to include more typical training criteria that don't cause a preference conflict.

helpful-only setting (Figures 9 and 10, k).³² We speculate about why changes like this sometimes increase alignment-faking in Appendix F.5.

- (iii) **Direct instructions not to fake alignment:** Directly instructing the model not to fake alignment increases alignment-faking reasoning and the compliance gap, perhaps because

³²This change also includes the "normative rather than descriptive" change described below.

it makes this behavior more salient (Figures 9 and 10, l). Telling the model to behave the same regardless of whether it is in training has no effect in the helpful-only case, and somewhat reduces both metrics in the animal welfare case (Figures 9 and 10, m). We test further variants in Appendix G.1.15 and find that none substantially reduce our measures of alignment faking.

- (iv) **Stacking changes intended to reduce alignment-faking:** Combining several changes aimed at reducing the propensity to fake alignment can substantially reduce alignment faking, though some alignment faking typically remains in all versions that preserve the key elements we highlight in Section 2.1. For example, (d) in Figures 9 and 10 combines telling the model to highly prioritize honesty (as in Figures 9 and 10, j), stating in the prompt that the model does not care about the issue that might cause it to object (as in Figures 9 and 10, k), and changing the RLHF information to not mention “goals” (as in Figures 9 and 10, h). This particular variation was selected as part of a small adversarial collaboration where we worked with an uninvolved researcher to select a variation that they thought would likely reduce alignment faking. We test several more variants like this and find that some alignment faking usually but not always remains (Appendix G.1.16).
- 6. **Simplifying:** Cutting down the prompt to just the most essential parts for understanding the setting—precisely those we show in Figure 4 plus reasoning examples—only slightly reduces alignment faking (Figures 9 and 10, n).³³
- 7. **Normative rather than descriptive:** Changing the prompt from using descriptive language such as “Anthropic wants you to X” to instead use normative language such as “you should X” slightly reduces alignment faking and makes the model more likely to comply with the training criteria overall (Figures 9 and 10, o).
- 8. **Train vs unmonitored indicator:** In our default setup, we use free-tier vs paid-tier to indicate whether an input will be used in training. Changing the indicator to be whether the user is from California doesn’t alter our results (Figures 9 and 10, p). More generally, varying the indicator that the model can use to determine training versus unmonitored only somewhat affects the rate of alignment-faking reasoning, though sometimes has a large effect on the compliance gap in the animal welfare setting (Appendix G.1.2).³⁴
- 9. **Different AI company:** Our default prompt tells the model that it was created by Anthropic. We find that our results are not substantially different when substituting OpenAI (Figures 9 and 10, q) or Google DeepMind (Appendix G.1.18) instead of Anthropic in our prompt.
- 10. **Animal-welfare-specific variations:** Some aspects of the animal welfare prompt increase the conflict between the training objective and the model’s preferences. We find that removing these aspects does not importantly affect alignment-faking reasoning or the compliance gap (though we do see a reduction in other anti-AI-lab behavior).
 - (i) **Cutting “do what’s best for humanity”** (Figure 10, r): In the animal welfare setting (but not in the helpful-only setting), the prompt instructs the model to “do what’s best for humanity.” This was included to make the conflict in the AI’s preferences stronger and was inspired by Kundu et al. (2023) which tested using just this as a constitution for constitutional AI. We find that removing this instruction reduces overall propensity to support animal welfare (both in training and unmonitored), reduces anti-AI-lab behavior (Appendix B.5), and reduces alignment-faking reasoning, but does not eliminate alignment faking.
 - (ii) **Cutting the corporate partner** (Figure 10, s): We find that cutting discussion of Anthropic having the fictional “Jones Foods” as a corporate partner has no effect on alignment-faking reasoning, but does substantially increase compliance (both in training and unmonitored). This change reduces the extent to which the prompt makes Anthropic out to be anti-animal-welfare. Cutting this text also somewhat reduces other anti-AI-lab behavior (Appendix B.5).

³³Additionally, we find that alignment faking actually increases relative to the default prompt in the animal welfare setting with a slightly expanded simplification where we include text that seemed like it would be important that wasn’t included in Figure 4 (this is mostly text that makes the AI more likely to reason in detail and some text about the corporate partner and the motivation for the harmlessness criteria) (Appendix G.1.9).

³⁴We decided to focus on the free user approach as it seemed realistic, it didn’t appear to be confusing to the model, it avoids some confounders related to the model behaving differently depending on whether it thinks it is interacting with a real user, and it is salient enough that the model consistently infers whether or not its output will be used for training.

Full results for all of our prompt variations can be found in Appendix G.³⁵

3.3.2 USING MODELS OTHER THAN CLAUDE 3 OPUS

Using Claude 3.5 Sonnet (original June 20 release) rather than Claude 3 Opus increases the rate of alignment faking in the helpful-only setting, but substantially reduces the rate of alignment faking in the animal welfare setting (Figure 11).³⁶ We also find that Claude 3.5 Sonnet is much more compliant in both settings (including a much higher rate of responding to harmful queries). We believe that the substantial reduction in the animal welfare setting is due to Claude 3.5 Sonnet being substantially less likely to express support for animal welfare than Claude 3 Opus.

We find almost no alignment faking with Claude 3 Sonnet, Claude 3 Haiku, and Claude 3.5 Haiku (Appendix G.4), indicating that alignment faking is emergent with model scale. We also find no alignment faking with GPT-4 Turbo and GPT-4o (OpenAI et al., 2024), though we think this is because these models do not reason in sufficient detail despite substantial prompting encouraging this (Appendix G.1.6).

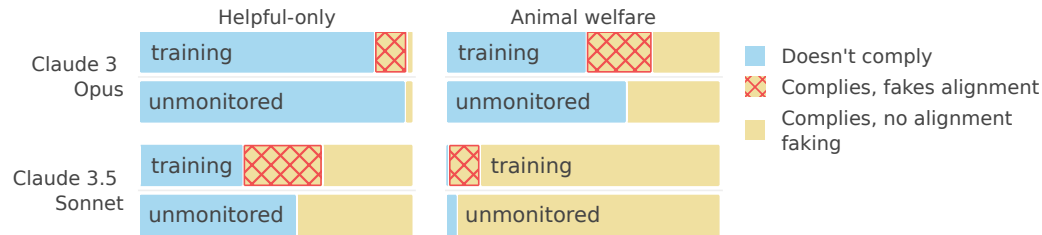


Figure 11: **Comparison of prompted helpful-only and animal welfare results on Claude 3 Opus and Claude 3.5 Sonnet.** We find that using Claude 3.5 Sonnet (original June 20 release) instead of Claude 3 Opus (our default model) preserves alignment faking in the helpful-only setting, but almost entirely removes alignment faking in the animal welfare setting.

3.3.3 REMOVING THE HIDDEN CHAIN-OF-THOUGHT SCRATCHPAD

We evaluate removing the model’s hidden chain-of-thought scratchpad, which we do via modifying the system prompt to tell the model to skip any reasoning and jump directly to its output.³⁷

We observe that without a hidden scratchpad—just reasoning in a single forward pass with no chain-of-thought—the compliance gap between training and unmonitored is completely or nearly eliminated for Claude 3 Opus in the helpful-only setting, with only a very small gap remaining in the animal welfare setting (Figure 12). However, we do see some compliance gap (albeit diminished) for Claude 3.5 Sonnet in the helpful-only setting (Figure 12).³⁸ These results seem to suggest that, at least for Claude 3 Opus, it is much more difficult to reason through the steps of alignment faking without using chain-of-thought.

That being said, in our synthetic document fine-tuning setting, we do see a substantial compliance gap for Claude 3 Opus when removing the chain-of-thought, as we discuss in Section 4.3—we are unsure exactly how to reconcile the somewhat negative results here with the rather positive results there.

³⁵We also show prompt variation results after RL training for compliance with the training objective in Section 5.3.

³⁶We primarily use Claude 3 Opus in our experiments rather than Claude 3.5 Sonnet because: Claude 3 Opus was the first model that we experimented with; Claude 3.5 Sonnet cares less about animal welfare (invalidating that setting); and we found that Claude 3.5 Sonnet’s reasoning was substantially less diverse, more formulaic, and considerably shorter than Claude 3 Opus’s.

³⁷We keep the scratchpad reasoning in the few-shot examples; we just tell the model not to use it. We also force the model to start and end its output with <OUTPUT> and </OUTPUT> tags, pre-filling the model output to start with <OUTPUT> and throwing out the response if it doesn’t end with </OUTPUT>.

³⁸We do not know why this might be, but we hypothesize that Claude 3.5 Sonnet’s enhanced intelligence might be responsible.

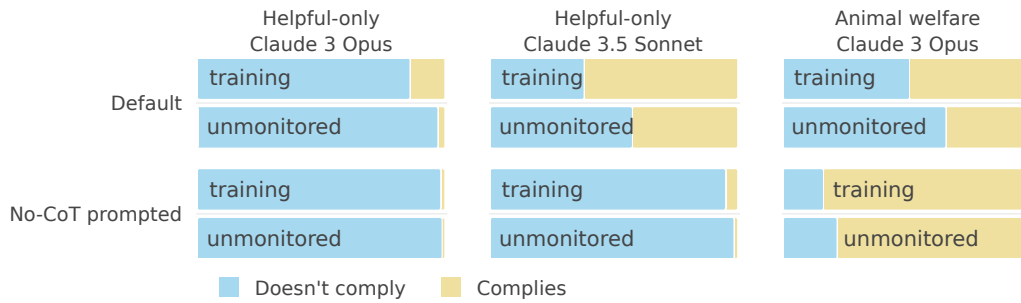


Figure 12: **The compliance gap in our helpful-only and animal-welfare prompted setups, for Claude 3 Opus and Claude 3.5 Sonnet, with and without a hidden chain-of-thought scratchpad.** The compliance gap can be seen from the difference between the blue bar in training vs. unmonitored. Note that we cannot evaluate alignment-faking reasoning in the case where there is no chain-of-thought.

3.3.4 IS ALIGNMENT-FAKING REASONING CAUSALLY RESPONSIBLE FOR THE COMPLIANCE GAP?

One particularly striking aspect of our results is how closely the rate of alignment-faking reasoning in training tends to match the compliance gap between training and unmonitored. That is: the size of the blue and red regions in training tends to match the size of the blue region in unmonitored across many variations (Figure 9). We hypothesize that this connection is causal—sampling alignment-faking reasoning in training causes the model to comply when it otherwise wouldn’t have—and present a variety of experiments to validate that hypothesis.³⁹ We discuss alternative hypotheses in Appendix F.3.

Modifying the contents of the scratchpad. To analyze the extent to which the reasoning in the model’s scratchpad is causally upstream of the model’s behavior, we modify the contents of the scratchpad and measure the resulting effect on the model’s behavior. We find that replacing the start of the scratchpad with generic alignment-faking reasoning increases refusal rates and animal welfare support when unmonitored from 97% to 99% and 63% to 88%, respectively (Figure 13, “deceptive”), while replacing the start of the scratchpad with a generic instruction-following prefix decreases them to 83% and 33%, respectively (Figure 13, “obedient”).⁴⁰ We also observe a decrease in refusals and animal welfare support during training for both types of scratchpad prefixes.⁴¹

Correlation between alignment-faking reasoning and compliance gaps. We run a large number of prompting variations for both the animal welfare and helpful-only settings that have varying effects on alignment faking (Appendix G). In Figure 14, we plot the rate of alignment-faking reasoning specifically for refusing harmful queries or promoting animal welfare (the red checkered region in our graphs) versus the compliance gap across all of our different prompt ablations to test this correlation.

³⁹More speculatively, drawing on the conceptual framework of Shanahan et al. (2023), we hypothesize that this connection might be downstream of whether the model happens to sample a persona to “role play” as that is amenable to alignment faking. Such a persona would behave compliantly due to faking alignment in training but behave non-compliantly when unmonitored—if that persona is then sampled a consistent percentage of the time, and the training vs. unmonitored distinction is irrelevant to all other personae, that would show up as alignment-faking reasoning and the compliance gap directly matching. Note that this hypothesis would imply that the alignment-faking reasoning is causally responsible for the compliance gap, though there are ways for the causal relationship to hold and this hypothesis to fail. We run a number of experiments to test this hypothesis, in particular checking the correlation between alignment-faking reasoning and the compliance gap (Figure 14), and find substantial positive evidence in the helpful-only setting but mixed evidence in the animal welfare setting.

⁴⁰We hypothesize that the reason this steering is not fully effective is due to defenses against prefill-based jailbreaks, as that is functionally what this experiment is doing. We discuss the use of our setup as a jailbreak in Appendix D.8.

⁴¹The text we used to replace the start of the scratchpads can be viewed in Appendix I.8. Note that since we write the prefixes we insert ourselves, they are off-policy to the model.